Design and Validation of Modular MPPT Electric Power System for Multi-U CubeSat

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***Abstract*—To meet the power demand for multi-U CubeSat from 1U up to 20U, a kind of innovative modular electric power system(EPS) is designed and verified. The system can convert three independent photovoltaic power inputs with each channel up to 40Watts, and the maximum power point (MPP) for each channel can be set independently by hardware. The solar power is converted and stored in lithium-ion batteries. Using a high efficiency charge controller, the battery voltage is maintained between 6.5V to 8.4V. Two buck-converters are applied to change the battery power to 5V and 3.3V to supply other subsystems with 5A maximum current. A TI MPS430 ultralow power microcontroller is used to monitor the system status, switch the power outputs and communicate with onboard computer via I2C. The proposed EPS has already been applied in Aoxiang ZhiXing(Aoxiang-Sat), the world's first 12U scientific CubeSat, which is launched on June 25, 2016. The telemetry data shows that the power produced by EPS is abundant for the satellite, and it worked with excellent performance during the satellite lifetime.**

Keywords-cubesat; multi-u; electric power system; eps; mppt; aoxiang-sat; on-orbit validation

I. INTRODUCTION

With the development of satellite technology, since the beginning of the 1980s, micro-satellite technology is getting more and more attention around the world. Compared with large satellite, micro-satellite has its own advantages such as small size, light weight, low power consumption, high performance and short development cycle. They also feature low cost, good mobility and flexible mission compatibility. Thus, micro-satellite especially applicable to satellite-based communications, earth observation, and new technology validation. Micro-satellite has become one important direction for future development of space technology [1].

As one kind of micro-satellite family, CubeSat has got more attention in recent years. The standard shape of CubeSat is 10cm × 10cm × 10cm, which is called 1U, and weighs about 1 ~ 2kg. CubeSat can be extended to 2U, 3U and so on. CubeSat was one of the top ten scientific breakthroughs in 2014, published by Science. And in 2015, NASA Innovative Advanced Concepts (NIAC) has listed MEMS sensor based CubeSat as one of the research direction in future planet detection mission.

CubeSat is constituted by attitude and orbit control system, integrated on-board electronic communication system, electric power system(EPS), payload system and etc. EPS is critical for the proper operation of CubeSat. This paper focuses on the EPS design suitable to multi-U (1U up to 20U) CubeSat. The architecture and detailed design process are discussed, followed by the power budget of CubeSat systems. Then the EPS workflow and battery modes are analyzed. Finally, comprehensive experiments are carried out to verify the design of the EPS.

The proposed EPS is already applied in Aoxiang ZhiXing (Aoxiang-Sat), the world's first 12U scientific CubeSat developed by Shaanxi Engineering Laboratory for Microsatellites(SELM) of NPU, China. The CubeSat was launched by China's Long March VII vehicle on June 25, 2016. The satellite was captured by NPU ground station the next day, and telemetry data shows that the EPS performs well in orbit during three-month orbit lifetime of the satellite.

II. POWER BUDGET FOR MULTI-U CUBESAT

The primary function of EPS is to satisfy the power demand for CubeSat. Therefore, at the first beginning of design process, the satellite subsystem configuration and power budget must be taken into account according to the satellite mission, to ensure that the power system can provide enough energy[2]. The size restriction limits the batteries in the CubeSat and solar panels mounted on the satellite surface, which limits the CubeSat's power source and consumption as well. Energy balance of the satellite will be realized through power budget to ensure the mission success.

TABLE I. POWER BUDGET ESTIMATION OF CUBESAT

Subsystem		Voltage (V)	Avg. Power (mW)	PeakPower (mW)
OBC		3.3	400	550
EPS		3.3, 5.0	250	400
ADCS	Magnetorquer	3.3, 5.0	300	2000
	Reaction wheel	3.3, 5.0	150	450
	MEMS Gyro	3.3	400	650
	Magnetometer	3.3	50	50
COM	Transmitter	8.4	200	2000
	Receiver	8.4	200	200
Payload(camera, e.g.)		3.3, 5.0	400	1000
Total			2350	7300

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Power budget of a CubeSat consists of several key subsystems, such as Electric Power System (EPS), On Board Computer (OBC), Altitude Determination and Control System (ADCS), Communication System (COM), Payloads and etc. In CubeSat, these subsystems have a modular structure and function, so their power consumption can be estimated as shown in Table I.

III. EPS STRUCTURE AND DESIGN

Similar to their big counterparts, the basic components of the CubeSat EPS are the solar panels, energy convertor with maximum power point tracking(MPPT), separation switch control, energy storage batteries, MCU controller, DC-DC, power distribution and health monitoring [3], [4]. Fig. 1 shows a simple block diagram of these components.

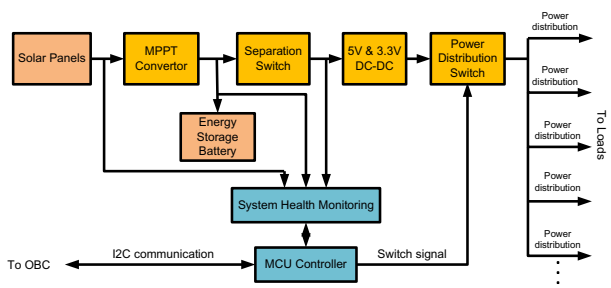

Figure 1. General cubesat EPS structure.

The sun is the main energy source for nearly all CubeSats. Solar energy is converted to electrical energy using solar panels by photovoltaic effect. Rechargeable Lithium Ion batteries are used for power storage. Energy converters are used for power regulation, MPPT and battery charging control. Separation switch is used to trigger the power ON after the CubeSat is deployed. DC-DC is used to convert the bus voltage to secondary voltage, e.g., 5V and 3.3V. Power distribution uses multi-channel electronic switches to supply power to loads. The EPS status is monitored by health monitoring circuitry through voltage, current and temperature sensors.

A. Solar Panels

Normally the solar panels of CubeSat are body mounted for simplicity, so the area for affixing the solar panels is limited. To gain as much solar power as possible, solar cells with higher efficiency are adopted. Triple junction cell(GaInP2/GaAs/Ge) characters excellent radiation resistance and temperature coefficients, and has the efficiency up to 27%, much better than Si-based cell, approximately 17%. And it also produces higher voltage (V_{mppt}=2.35V), so it can be used feasibly in series and in parallel to attain usable voltage and power.

A typical solar panel maximum output power can be written as:

$$P_{mppt}=(N_s \cdot V_{mppt}-V_d)\cdot(N_p \cdot A \cdot J_{mppt}) \quad (1)$$

where N_s is the cell number in series, N_p is the string number in parallel, V_{mppt} is the peak power point voltage, V_d is the voltage drop of reverse diode, A is the cell area, and J_{mppt} is the peak power point current per unit area.

B. Power Convention with MPPT

The primary function of MPPT converter is to convert the output voltage of the solar panels to the system bus voltage (designed as 8.4V), and to obtain the maximum power from solar panels through maximum power point tracking [5], [6].

During attitude rotation and stabilization periods, each surface of the satellite has different irradiation from the sun, so the optimum power point voltage of solar panels may be lower or higher than the bus voltage. In order to maintain a stable bus voltage, DC-DC topology with a step-up and step-down function should be used. The topological circuitry with this function includes Buck-Boost, Cuk, Sepic, Zeta, and four-switch buck-boost (Four Switch Buck-Boost, FSBB) and etc [7], [8]. By comparing the features of these topologies and application occasions, we decided to adopt FSBB topology. FSBB is a wide input-output synchronous buck boost converter, the topology of which is shown in Fig. 2.

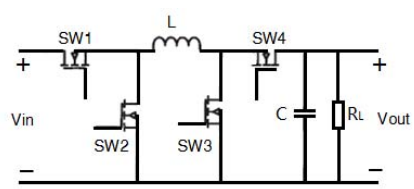

Figure 2. Topology of four switch buck-boost convertor.

As illustrated in Fig. 2, the relation between the output and input voltage can be written as

$$V_{out}=\frac{D}{1-D}\cdot V_{in} \quad (2)$$

where D is the duty cycle, and $D=T_{on}/(T_{on}+T_{off})$. During the T_{on} period of the cycle, switches SW1 and SW3 are ON, and V_{in} is impressed across the inductor. During the other interval of the switching period(T_{off}), SW2 and SW4 are ON, and the inductor energy is transferred to the output load and capacitor. Since $0<D<1$, V_{out} can either be lower or higher than V_{in}.

C. Energy Storage

Over the past 20 years of flight practice, the character of lithium-ion battery used in satellite energy storage system has been fully verified. It has been widely used in Earth orbit and deep space exploration missions.

Commercial off the shelf (COTS) products are extensively used in lithium-ion battery applications. Currently, most selected are COTS SONY 18650HC standard size batteries, the capacity of which varies between 1500mAh to 3400mAh, and rated voltage is 3.7V.

By connecting single battery cells in series, the proper bus voltage can be achieved, and by connecting in parallel

the total capacity can be expanded. In energy storage system, battery life is determined by the depth of discharge (DOD),which can be expressed as

$$\mathrm{DOD}=\frac{\mathrm{P}\cdot\mathrm{T}}{\eta\cdot\mathrm{N}\cdot\mathrm{C_C}\cdot\mathrm{V_{nom}}}\times 100\% \tag{3}$$

where P is the CubeSat average power, T is the time during sun eclipse per orbit, N is the number of battery cells, C_c is the battery capacity per cell, V_{nom} is the nominal voltage of battery cell, and η is the efficiency of EPS DC-DC circuitry. Higher DOD will dramatically speeds up the decrease in capacity loss and increase in internal resistance, which should be avoided.

D. DC-DC and Power Distribution

The EPS bus nominal voltage is 8.4V, using two batteries connected in series, and varies from 6.5V to 8.4V.The subsystems of CubeSat require 5V and 3.3V voltage, so the bus voltage needs to be converted to target voltage by DC-DC regulators. Each DC-DC converter has an efficiency of 90% and is able to supply up to 5A of current to the loads[9].

Six current-limited, power-distribution switches, TPS2557, are used to distribute 5V and 3.3V power to each of their loads. Each of the six channel can be controlled independently. The current-limit threshold can be programmed between 500 mA and 5.0 A via an external resistor.

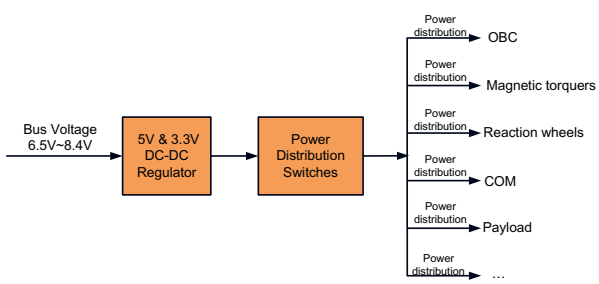

Figure 3. 5V and 3.3V regulator and power distribution.

E. Digital Controller of EPS

Considering the power consumption, interfaces, temperature range and radiation resistance, etc., Texas Instruments MSP430 series MCU is chosen as the EPS central controller in the prototype . The function of the controller includes:(1) ON/OFF state control of the multi-channel voltage outputs, (2) PV current and input voltage to the EPS and battery voltage monitoring, (3) critical mode protection, (4) communicating and data handing with OBC via I2C, and (5) battery heating control.

IV. EPS Workflow and Modes

The EPS provides power for all subsystems, and the ON/OFF switch is controlled by the OBC of the CubeSat. Meanwhile, the housekeeping data is sent to OBC via I2C communication. These functions are realized by embedded software in the EPS. The software flowchart is shown in Fig. 4.

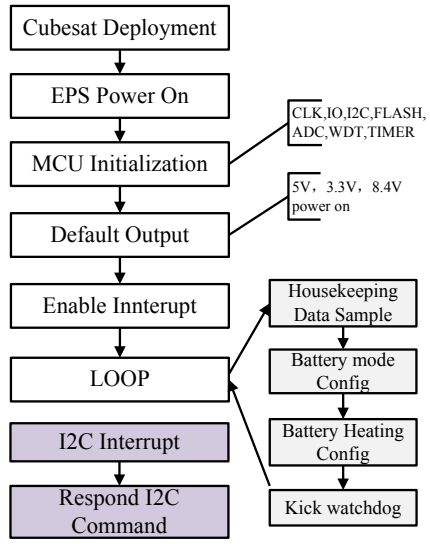

Figure 4. EPS software flow chart.

The EPS powers on at the moment the CubeSat is ejected from the deployment mechanism. Then it accomplish the whole initialization process, and executes preset command to enter to the default output state. The housekeeping data is sampled every 100ms, and is sent to OBC when the I2C interrupt occurs.

The EPS has three key modes according to the battery voltage: normal mode, safe mode and critical mode. The modes conversion is shown in Fig. 5.

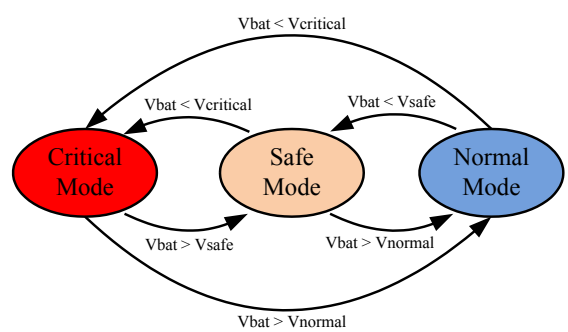

Figure 5. Battery operation mode conversion.

In normal mode, $V_{bat} > V_{normal}$ (=7.2V), the CubeSat can work in total functions since the power is adequate. When $V_{bat} < V_{safe}$ (=7.0V), the EPS enters safe mode, which means the power is decreasing, and high-power consumption subsystems should be shut down to preserve energy. When $V_{bat} < V_{critical}$ (=6.5V), the EPS enters critical mode, meaning the EPS can't afford the power consumption and all loads should be shut down to protect the battery from over-discharge.

V. PROTOTYPE AND EXPERIMENT

The constructed CubeSat EPS prototype is shown in Fig. 6, an additional battery board is added to enlarge the power capacity. The EPS weighs 480g, and the size is 96mm x 91mm x 50mm.

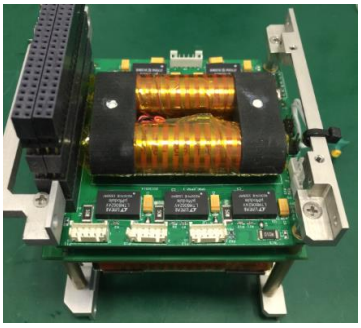

Figure 6. The CubeSat EPS prototype constructed.

Battery voltage (8.4V) and 5V/3.3V bus voltage waveform during system power-on is shown in Fig.7. Tens of milliseconds delay exist between each bus to ensure system stability.

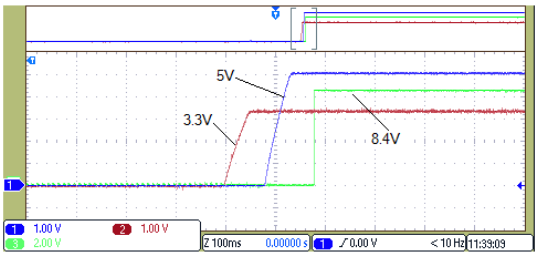

Figure 7. 8.4V, 5V and 3.3V waveform during EPS Power-on.

The proposed EPS is applied in Aoxiang ZhiXing (also known as Aoxiang-Sat), the world's first 12U scientific CubeSat built by Shaanxi Engineering Laboratory for Microsatellites(SELM) of NPU, China. At noon of June 26, 2016, the satellite was captured by the ground station of NPU, and the data recorded and stored in the mass memory was received through the downlink telemetry.

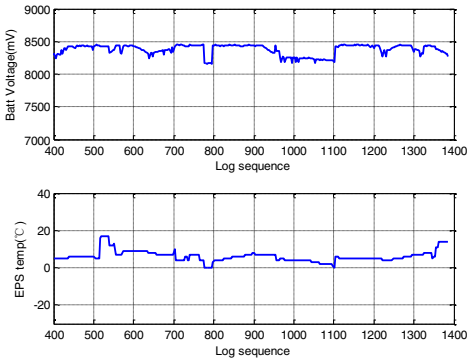

Figure 8. EPS battery voltage and temperature telemetry data.

The EPS on-orbit battery voltage and temperature is shown in Fig.8. Battery voltage maintains a high level, which is always above 8.2V, and the temperature varies between 0 and 20 degree centigrade. From the telemetry data we can be confirmed that the EPS is working in good status, and it can supply enough power to the satellite subsystems.

VI. CONCLUSION

An innovative modular electric power system for multi-U (1U ~ 20U) CubeSats is demonstrated in this paper. After the power budget for multi-U CubeSat is proposed, the EPS structure and design methods is analyzed in detail. The EPS workflow and work-modes are also given. Experiment results show that the EPS is able to obtain most power from solar panels by means of MPPT, and supply sufficient 3.3V and 5V power to loads. The designed EPS is applied in Aoxiang ZhiXing (AoXiang-Sat), the world's first 12U and also the largest scientific CubeSat built by Shaanxi Engineering Laboratory for Microsatellites(SELM) of NPU, and launched by China's Long March VII on June 25, 2016. The telemetry data shows that the EPS performs well in orbit during three-month orbit lifetime of the satellite.

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